Tutorial # 1.

Machine , Simulations

Exercise 1. Warming up

Q1. Write the 0 letters of the word on a 2nd tape.
Write the 1 letters of the word on a 3rd tape.
Compare the length of the words on the 2nd & 3rd tapes

Q2. Replace the # by a 1, remove the last 1.
Given a string w on the tape, write the last part on a 2nd tape.
Then compute addition and compare the results.

Q3. Write a and b on the 2nd & 3rd tape.
Until the start of a is reached, step to the left (2nd tape)
Write $\langle 1^{st}$ tape$\rangle$ # $\langle 3^{rd}$ tape$\rangle$ on the 1st tape
and compute addition.

Exercise 2. Universal Turing Machines

Q1.

tape #1	a_1^1	a_2^1	a_3^1	...
tape #2	a_1^2	a_2^2	a_3^2	...
⋮	⋮	⋮	⋮	... ⋮
tape #k	a_1^k	a_2^k	a_3^k	...

where we encode each letter in binary

$\rightsquigarrow$

$\triangleright a_1^1$	$\mathcal{B} a_1^1$	...	$\triangleright a_1^k$	$\mathcal{B}\ a_2^k$	...

and we have a tape whose size is at most

$$O(\log|\Gamma|) \times 2k \times s(|x|).$$
$$\overset{\shortparallel}{O(s(|x|))}.$$

Q2. Use a tape for $\langle \alpha, x\rangle$
and a tape for the state of M_α
and 2 tapes for the tapes of M_α. $\Big\}\ C_\alpha$

$$\#\text{ used cells} \leq C_\alpha \times s(|x|)$$

Exercise 3. Stimulating Simulation.

Q1. $\cdots \mid a_{-3} \mid a_{-2} \mid a_{-1} \mid a_0 \mid a_1 \mid a_2 \cdots \rightsquigarrow \cdots \mid a_{-3} \mid a_{-2} \mid a_{-1} \mid a_0 \mid a_1 \mid a_2 \cdots$

Q2. Write each letter of Γ in binary, the cost is $\lceil \log_2 |\Gamma| \rceil$.

Q3. Store the two tapes on an alphabet $\Gamma^2 \times \{B, \triangleright\}$.
We don't talk about the heads of both tapes.
Reduce the size of the alphabet w/ Q2.

Q4.

TD n° 2.

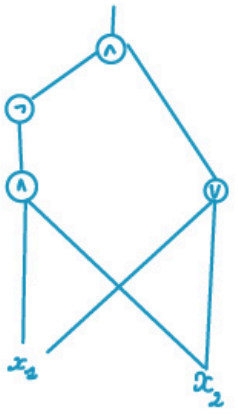
The class NP

Exercise 1. Boolean circuits.

1) f_0 :

f_1 :

f_2 :



2) We have: $f_{2n}(\vec{x},\vec{y}) = f_2(f_n(\vec{x}), f_n(\vec{y}))$

and thus we have a size $O(n)$ and $O(\log n)$ depth.

Exercise 2. One-way function.

1) Assume $P = NP$ and that f is a one-way function. Consider n.
As f is poly-time computable, there exists a circuit of poly-size computing f. As Circuit SAT $\in NP = P$, we can get an antecedant to $f(x) \in \{0,1\}^s$ (with $s = poly(n)$) by the circuit in time $poly(n)$.

Thus $F(-, 1^n)$ is computable in poly-time and gives an antecedant with probability 1.

Absurd as f is a one-way function.

2)

Exercise 3. NP-completeness.

1) Consider a problem $R \in NP$. Then it follows that a poly-time verifier V exist.

We will show that $R \leq_p$ NDTM-T. Consider w an input of R and

we have that

$$w \in R^+ \iff V(w, \underset{\text{certificate}}{_}) \text{ accepts some input in time } \leq Q(w)$$

where Q is a polynomial bound on the runtime of V.

This problem is NP by simulating M_α on input x until time t.

TD n° 4

More Space

Exercise 1. NL-completeness.

Q1. $L \subseteq NL$

$NL \subseteq DTIME(\exp(\log n)) \subseteq P$.

Q2. Let $f : A \leq_\ell B$ and $g : B \leq_\ell C$ then $g \circ f : A \leq_\ell C$ as

$$x \in A \iff f(x) \in B \iff g(f(x)) \in C$$

and $g \circ f$ is computable in log space as $|g(x)| \leq \exp(\log |x|)$
$\leq poly(|x|)$

If $f : A \leq_\ell B$ and $B \in NL$ then we can simply check membership of x in A by computing $f(x)$ and checking membership of $f(x)$ in B, this can be done in log space.

Q3. No as we'd have that every language $A \in P$ in NL.

Q4. Consider a language $A \in NL$ and M a non-deterministic program running in log space.

Then consider the graph G of all executions of M with the initial configuration x and all the accepting configurations of M as $y_1, \dots, y_n$.

Add a node y and connect $y_i \to y$.

Then M accepts $x \iff$ there is a path from x to y in G

and the construction of G can be done in log space.

Q5. Strong Conn is in NL and Strong Conn $\leq_\ell$ Path.

Consider a graph G and add edges from every vertex $\neq x, y$ to x and from y to each vertex $\neq x, y$.

We have that $\tilde{G}$ is strongly connex iff there is a path $x \rightsquigarrow y$ in G.

Exercise 2. Little space is no space at all.

Q1. If there exists $k \in \mathbb{N}$, for all $x \in \mathcal{L}$, $M(x)$ uses less than k cells on the work tape, then $\mathcal{L} \in$ DSPACE(1).

If $\lim_{k \to \infty} \frac{|x|}{k} \to \ell$ then $\mathcal{L} \in$ DSPACE(1).

Q2. There is at most $|Q|$ states and at most $2^{O(\log \log n)} = o(\log n)$ work tapes with $O(\log \log n)$ possible work head positions.

Q3. All even i lead to different crossing sequences thus we have $O(|x|)$ pairwise distinct sequences.

Q4.

Exercise 3. Deterministic time hierarchy.

Q1. Consider (α, x) and let us test membership in $\mathcal{L}_g$ in time $O(g(|(\alpha,x)|))$ by simulating $g(|x,\alpha|)$ steps of $\mathcal{U}(\alpha,(\alpha,x))$, and saying if it is rejected.

TD n°5

Exercise 1: Deterministic time hierarchy.

Q1. c.f. TD n°4

Q2. If $\mathcal{L}ang(M_\alpha) = \mathcal{L}_g$, then consider $\langle \alpha, w \rangle$:

If $\langle \alpha, w \rangle \in \mathcal{L}_g$, then $\mathcal{U}(\alpha, \langle \alpha, w \rangle)$ will run for at most $g(|\alpha, x|)$ and rejects.
$\qquad\qquad\qquad

Exercise 2. Gap theorem.

flemme...

Exercise 3. Non-deterministic time hierarchy.

Q1. No as we have multiple executions that need to halt and reject all at once.

Q2. It's in $O(2^{g(n)})$.

Q3. If $i \geq 2^{g(m)}$ then we take $O(i + g(n)) = O(g(n))$ time.
Otherwise $i < 2^{g(m)}$ then it'll run in $O(g(n) + g(n)) = O(g(n))$

Q4. Let, by contradiction, N such that $N(\langle \alpha, 1^i, x \rangle) = V(\langle \alpha, 1^i, x \rangle)$
then $N(\langle \alpha, 1^i, x \rangle)$ accepts iff $\begin{cases} N_\alpha(\langle \alpha, 1^i, x \rangle) \text{ rejects} \\ U_{ND}(\langle \alpha, \langle \alpha, 1^{i+1}, x \rangle) \text{ rejects.} \end{cases}$

$\overset{\shortparallel}{N_\alpha(\langle \alpha, 1^{i+1}, x \rangle)}$

and runs in time $\leq c f(n+1) = O(g(n))$.

Thus $NTime(f(n)) \subsetneq NTime(g(n))$.

Exercise 4. $P \neq DSpace(n)$.

Q1. We copy x on the tape and check that the input is of the form $\langle \underbrace{x, 1^{|x|^2}}_{\text{length } n = m^3} \rangle$ as $|x|^2 = O(n)$.

Then we check if $x \in L$ in $O(m)$ space.

Q2. If $P = DSPACE(n)$, we show that $L \in P$ using that $L_{square} \in P$. To do that, we simply construct $\langle x, 1^{|x|^2} \rangle$ in poly time and check membership in L_{square}.

Thus $L \in P = DSpace(n)$.

Q3. If $P = DSPACE(n)$ then we have $DSpace(n^2) \subseteq DSpace(n)$.

$$\supseteq$$

$$\overset{\uparrow}{=}$$

Absurd by space hierarchy.

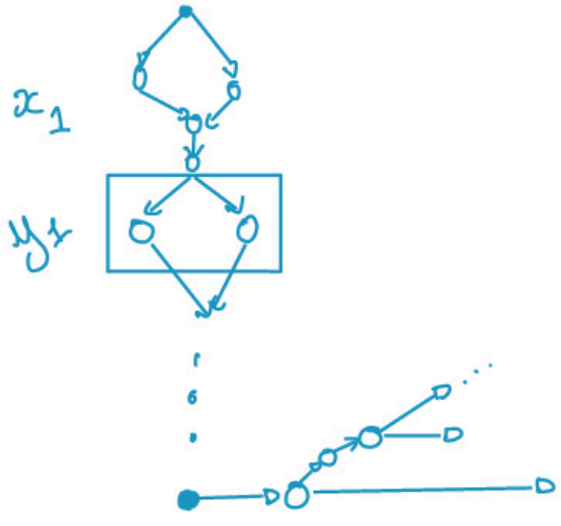
Exercise 5. Generalized Geography.

Q1. Player 1 gets to choose the value of x in a $\exists x \ldots$

Player 2 gets to choose the value of x in a $\forall x \ldots$

Q2. Consider $\varphi = \forall x_1 \exists y_1 \cdots \forall x_n \exists y_n \bigwedge_{i=1}^{m} \bigvee_{j=1}^{k} l_{ij}$

where $l_{ij} = x_n$ or $\overline{x}_n$.



clause $x_1 \vee \overline{x}_2 \vee x_3$

Tutorial n° 6

Exercise 1.

Q1. If M runs in time $f(n)$ and uses an oracle for $A \in P$.
Then let N run in time $g(n)$ and recognize A.
Then we can simulate M and run in time

$$f(n) \times g(n).$$

This way, $P^P \subseteq P$, $NP^P \subseteq NP$

$$EXP^P \subseteq EXP \quad \text{and} \quad NEXP^P \subseteq NEXP.$$

The reverse inclusions are obviously true.

Q2. Let $EXPCOM := \{\langle M, x, 1^n \rangle \mid M \text{ accepts on } x \text{ in} \leq 2^n \text{ steps}\}$.

We have that $P^{EXPCOM} \subseteq NP^{EXPCOM} \subseteq EXP$.

Furthermore, let M that runs in time 2^n, then consider

1) ask the oracle if $\langle M, x, 1^n \rangle \in_{EXPCOM}$ (constructible in $O(n)$)

thus $EXP \subseteq P^{EXPCOM}$.

Q3. By Q1.

Q4. Negate the output of the oracle.

Q5. Let M that runs in (non-det) time 2^n. Then the language

$$\{\langle x, 1^{2^n} \rangle \mid x \in L(M)\}$$

is in NP.

Q6. If $NP = coNP$, then we can decide every language in NP, and thus $NP = NP^{NP}$.

Conversely, if $NP^{NP} = NP$, then we can solve an $coNP$ problem Q in NP by calling an oracle for $\overline{Q} \in NP$.

Q7. If $EXP = NP$ then $coEXP = coNP$, and then

$$\begin{array}{c} coEXP = coNP \\ \| \\ EXP \\ \| \\ NP \end{array}$$

$$NP^{SAT} = NP^{NP} = NP \subsetneq NEXP \underset{Q5}{\subseteq} EXP^{NP} = EXP^{SAT}$$

Exercise 2.

Q1. A tally language is fully represented by a subset of $\mathbb{N}$. Use χ_A for $A \subseteq \mathbb{N}$ as hint.

Q2. Encode the halting problem in unary.

Q3.